

Algebra General Exam - January 2023

Your UVa ID Number:

- Please Prove all your work and justify any statements that you make.
- State clearly and fully any theorem you use.
- Vague statements and hand-waving arguments will not be appreciated.
- You may assume the statement in an earlier part Proven in order to do a later part. do EACH PROBLEM ON A SEPARATE SHEET OF PAPER, AND STAPLE THEM TOGETHER IN THE CORRECT ORDER BEFORE TURNING THE EXAM IN.

Sign below the pledge:

"On my honor, I pledge that I have neither given nor received help on this assignment."

Problem 1

(a) (8 points) Let G be a finite group, and $H \subsetneq G$ be a proper subgroup. Prove that

$$G \neq \bigcup_{g \in G} gHg^{-1}$$

(Hint. Estimate the number of elements in the right-hand side.)

(b) (8 points) Let G be a transitive subgroup of the symmetric group S_n . Prove that there exists $g \in G$ that has no fixed points on $I_n = \{1, 2, \dots, n\}$ (i.e. there is no $i \in I_n$ such that $g(i) = i$).

Problem 2

Let $R = \mathbb{Z}[\sqrt[3]{2}] = \{a_0 + a_1\sqrt[3]{2} + a_2(\sqrt[3]{2})^2 \mid a_0, a_1, a_2 \in \mathbb{Z}\}$.

(a) (5 points) Prove that R is a subring of $\mathbb{R}$.

(b) (6 points) Prove that the evaluation homomorphism $\mathbb{Z}[x] \rightarrow R, p(x) \mapsto p(\sqrt[3]{2})$, yields a ring isomorphism

$$R \simeq \mathbb{Z}[x] / (x^3 - 2)$$

(c) (6 points) Using the isomorphism from part (b) and the 3rd Isomorphism Theorem, can you determine if the ideals $5R$ and $7R$ are prime or maximal?

Problem 3

(8 points) Let F be a field (possibly finite). Prove that the polynomial ring $F[x]$ has infinitely many prime ideals.

Note: You cannot assume without proof that $F[x]$ has infinitely many irreducible polynomials.

Problem 4

(10 points) Let R be a commutative ring with 1. Let $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ be a $\mathbb{Z}$ -module homomorphism, and let $K = \ker f$. Prove that there exists a submodule $M \subset \mathbb{Z}^n$ such that $\mathbb{Z}^n = K \oplus M$. (Hint. Argue that $P := \operatorname{im} f \subset \mathbb{Z}^m$ is projective.) Reminder. An R -module P is called projective if, for any surjective module homomorphism $\varphi : M \rightarrow N$, every module homomorphism $\psi : P \rightarrow N$ admits a lifting $\tilde{\psi} : P \rightarrow M$ such that $\varphi \circ \tilde{\psi} = \psi$.

Problem 5

Let L/K be a finite Galois extension of degree d with Galois group G . In this problem, we think of the elements of G as linear transformations $\sigma : L \rightarrow L$ of the d -dimensional vector space L over K .

(a) (5 points) Let $\sigma \in G$ be an element of order n . Prove that the minimal polynomial of $\sigma : L \rightarrow L$ is $\mu(x) = x^n - 1$. (Hint. Use the standard theorem about linear independence of characters.)

(b) (10 points) Assume that p is a prime $\neq \operatorname{char} K$, and K contains a primitive p -th root of unity. Let L/K be a Galois extension of degree p , and let $\sigma \in \operatorname{Gal}(L/K)$ be a nontrivial automorphism. Prove that σ is diagonalizable. Identify its eigenvalues, the corresponding eigenvectors and derive from this analysis that $L = K(\sqrt[p]{a})$ for some $a \in K^\times$. (Note. This gives an easier proof of Kummer theory in this special case).

Problem 6

(10 points) Let $K/\mathbb{Q}$ be a Galois extension with Galois group S_n (symmetric group) for some $n \geq 5$. Assume that K contains a primitive root of unity ζ_d . Prove that $d \leq 6$.

Problem 7

(a) (6 points) Let L/K be a field extension and $f(x) \in K[x]$. Prove that there is an isomorphism of L -algebras

$$L \otimes_K \frac{K[x]}{f(x)K[x]} \simeq \frac{L[x]}{f(x)L[x]}$$

Note. The L -algebra structure on the ring $L \otimes_K \frac{K[x]}{f(x)K[x]}$ is given on simple tensors by

$c \cdot (a \otimes \overline{g(x)}) := (ca) \otimes \overline{g(x)}$, where ca is the ring multiplication in L .

(b) (7 points) Suppose that L / K is a finite Galois extension of degree n . Prove that there is an isomorphism of L -algebras

$$L \otimes_K L \simeq \underbrace{L \times \cdots \times L}_n.$$

Problem 8

Let V be the $\mathbb{C}$ -vector space of all polynomials $p(x, y)$ in two variables of total degree ≤ 2 . Let $T : V \rightarrow V$ be the linear transformation $T(f) = \frac{\partial f}{\partial x}$.

(a) (2 points) Prove that T is nilpotent.

(b) (9 points) Construct a Jordan basis for T and compute its Jordan canonical form.